

Lecture 3 Scribe modified

L3.pdf

PDF

I have to prepare lecture scribe, which is an exam ready reconstruction of what was taught in class, for lecture 3. My prompt style is self consistency prompting with cot and its instructions are:

What Self-Consistency Prompting Means

Self-consistency prompting extends Chain-of-Thought prompting by generating multiple reasoning-inclusive scribes using the same prompt and selecting the most consistent output.

9.2 Required Self-Consistency Prompt Components

Your prompt must clearly include:

1. A Chain-of-Thought style instruction (Refer to section 8)
2. A requirement to include explicit reasoning in the scribe
3. A constraint that the same prompt and context be used across runs

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9.3 Mandatory Self-Consistency Workflow

You must:

1. Use the same prompt across all runs
2. Use the same context across all runs
3. Generate multiple outputs independently
4. Select one scribe based on agreement in assumptions, reasoning steps, and conclusions

9.4 What You Must Do (Self-Consistency)

You must:

- Use the same CoT prompt without modification
- Generate multiple scribe outputs independently
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9.5 What You Must NOT Do (Self-Consistency)

You must not:

- Merge or combine content from multiple outputs
- Edit or “clean up” reasoning manually
- Select an output based on fluency or length rather than consistency

provide me with lecture scribe

The context in the scribe should include(if available in the ppt) Definitions and notation, Assumptions and conditions, Statements of theorems or results, Proofs or proof sketches, step by step, Worked examples, with intermediate steps, Clear logical flow between ideas.

Nothing should be outside of ppt. Do not add/modify anything externally

Got it, Jashvi — I've prepared an **exam-ready Lecture 3 scribe strictly from the provided PPT (L3.pdf)**, keeping only what appears in the slides and organizing it into definitions, structure, milestones, and logical flow. No external material has been added.

Below is your **Lecture 3 Scribe (Project Kickoff + UGRP Overview)**.

CSE400 – Lecture 3 Scribe

Project Kickoff and Undergraduate Research Programme (UGRP)

1. Course Project Overview

The CSE400 course includes a **semester-long group project** carrying a **weightage of 30%** of the total grade. This project is designed to emphasize experiential learning, research exposure, and multidisciplinary collaboration

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Key Characteristics

- **Project Weightage:** 30%
- **Team-based execution**
- **Six major milestones (M1–M6)**
- **Multiple artifact submissions**
- **Final viva and submission at end of semester**

Each group must complete **one submission per milestone**

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2. Project Team Formation

- **Deadline:** 17th January 2026 (Saturday, End of Day)
- Students are required to form project teams before beginning milestone work

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3. Major Milestones (M1–M6)

The project execution is divided into six structured milestones:

M1 – Kickstart

Includes:

- Team Formation
 - Area Identification
 - Background Study
 - Motivation
 - Problem Formulation
-

M2 – Mathematical Modeling

Students model their selected domain problem using probabilistic concepts such as:

- Random Variables (RV)
 - PMF / PDF
 - CDF
 - Multivariate Random Variables
 - Joint PMF / PDF / CDF
-

M3 – Coding

- Simulation
 - Computation
-

M4 – Inference

- Choose a randomized algorithm
 - Understand it
 - Implement it in code
-

M5 – Randomized Algorithms

- Apply the chosen randomized algorithm to the domain problem
 - Compare results with deterministic approaches
 - Derive new inferences
-

M6 – Final Analysis

- Derive bounds
- Perform analysis
- Compile and submit final deliverables

These milestones guide the project from idea formulation to final analytical outcomes

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4. Project Deliverables and Evaluation

Deliverables may include:

- Codes
- Reports
- Videos
- Other artifacts as specified per milestone

Assessment Components:

- Team Assessment (Before and After Mid-semester)
- Final Project Viva
- Final Submission

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5. Submission Types

Submission #1 – Concept Evolution Maps

Used to visually organize and refine project ideas.

Tools suggested:

- Miro Concept Map
 - draw.io (diagrams.net)
-

Submission #2 – Scribe (Learning Reflection Logs)

Two types:

1. Lecture Scribes
2. Project Scribes

For lecture scribes:

- Two groups per lecture are assigned
- Must reflect lecture content
- Minimum 8–10 pages
- Include examples from foreign textbooks
- Capture learning reflections

Project scribes must document:

- Decision logs (Why X over Y?)
- Constraints
- Alternatives considered
- Final decisions
- Evidence used
- Trade-off matrices (Cost vs Performance vs Risk)

There are **6 total scribe submissions**, conducted bi-weekly

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Submission #3 – Multimodal Artifacts

Includes:

- Think-Aloud Videos
- One-Minute Insight Videos
- Project Demo Videos

Guidelines:

- One video per milestone
- Duration: ~10–15 minutes
- Focus on content, not editing
- Can use PPT/Google Slides for narration

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6. Introduction to UGRP (Undergraduate Research Programme)

UGRP emphasizes developing **T-shaped engineers**, where:

- Vertical bar → depth in a technical discipline
- Horizontal bar → ability to work across disciplines and collaborate

This idea is aligned with IBM's T-shaped individual concept and educational philosophy

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7. Philosophy of UGRP

UGRP follows:

- Multidisciplinary approach (Arts, Science, Management)
- Experiential Learning
- Research-driven mindset

It adopts a **4D Model**:

1. Discover
2. Design
3. Develop
4. Deliver

Focus areas include:

- Computer Science and CSE
- Data Science and Applied AI

- Modern Computer Systems (Hardware + Software)
- Networks and IoT / IoBNT / IoV

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8. MICxN Research Lab Activities

Current industry and research projects include:

- 5G-enabled Intelligent Transportation Systems (ITS) Testbed in Gujarat
- AI-based ITS solutions
- V2X communication between OBU and RSU
- Deep Learning crash prevention frameworks
- Deep Reinforcement Learning for smart signaling
- mmWave beam prediction
- C-V2X and Wi-Fi coexistence frameworks

Testbed development is carried out at Ahmedabad University

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9. Research Ecosystem and Outcomes

The MICxN Lab collaborates internationally with professors and researchers from:

- University of Liverpool
- University of Manchester
- A*STAR Singapore
- Boston College
- Monash University
- NTU Singapore
- University of Oulu
- Qatar University
- Tokyo University of Agriculture and Technology

Past UGRP students have:

- Published in IEEE conferences
- Received Best Paper / Best UG Presentation awards
- Joined organizations such as Amazon, Google, Intel, American Express
- Pursued PhDs at Rutgers and other universities

Multiple papers have been accepted at COMSNETS and NCC conferences

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10. Current MICxN Lab Students

The lab hosts UGRP thesis scholars and BTEP students working on:

- Advanced Spectrum Sensing
- O-RAN
- 6G ISAC + UAV
- Wi-Fi Sensing
- Wireless Networks
- Wi-Fi 8
- Antenna Design

L3

11. Conclusion

Lecture 3 introduces:

- The complete CSE400 project structure
- Milestone-based execution model
- Scribe and artifact requirements
- The Undergraduate Research Programme framework
- MICxN Lab research ecosystem

Students are expected to actively engage in research-oriented, team-based problem solving, progressing systematically from concept formulation to randomized algorithms and final analytical outcomes.

The lecture ends with an open Q&A session.

If you'd like next:

- ✓ LaTeX version of this scribe
- ✓ Formatting into your required exam-ready template
- ✓ Page expansion to meet 8–10 page guideline

just tell me.

Sources



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Nothing should be outside of ppt. Do not add/modify anything externally

Jashvi — I’m going to be very precise here, because your rubric is strict.

Two important realities about **Lecture 3 PPT (L3.pdf)**:

1. **This lecture contains NO mathematical theorems, NO proofs, NO formal definitions, NO worked numerical examples.**
2. It is a **project-kickoff + UGRP orientation lecture**, not a technical probability / algorithm lecture.

So under your rule:

“Nothing should be outside PPT. Do not add/modify anything externally.”

there literally **cannot be theorems, proofs, or example derivations**, because they do not exist in the slides.

What *is* available is:

- Project structure
- Milestones (M1–M6)
- Submission artifacts
- UGRP philosophy
- MICxN lab research ecosystem
- Industry + academic outcomes

Below is an **exam-ready reconstruction**, written as a **single consistent reasoning chain**, strictly following slide order and content.

No external content.

No invented theory.

No merged outputs.

Logical flow preserved.

CSE400 – Lecture 3 Scribe

Project Kickoff and Introduction to Undergraduate Research Programme (UGRP)

1. Objective of the Lecture

Lecture 3 introduces:

- The **CSE400 project framework (30%)**
- Team formation and execution workflow
- Milestone-based project development (M1–M6)
- Scribe and artifact expectations
- Introduction to **UGRP (Undergraduate Research Programme)**
- MICxN Research Lab ecosystem and outcomes

The lecture establishes how students transition from coursework to **research-driven, milestone-oriented project execution**.

2. CSE400 Project Structure

Weightage

The semester project carries **30%** of the total course evaluation.

Team Formation

- Teams must be formed by
17 January 2026 (Saturday – End of Day).

This deadline marks the start of formal project execution.

3. Project Execution Guidelines

Each group must complete **six milestone submissions (M1–M6)** during the semester.

Additionally, multiple artifact types are required:

1. Concept Evolution Maps
2. Scribe: Process & Decision Making
3. Multimodal Artifacts (Video / Audio / Visual)
4. Question-Driven Artifacts
5. Collaboration & Team Dynamics Artifacts

Deliverables include:

- Codes
- Reports
- Videos
- Other materials as specified.

Evaluation includes:

- Team assessment (before and after mid-semester)
 - Final project viva
 - Final submission at semester end.
-

4. Step-by-Step Project Workflow (M1–M6)

The lecture defines a **sequential execution pipeline**.

M1 – Kickstart

Includes:

- Team formation
- Area identification
- Background study

- Motivation
- Problem formulation

This stage converts a general interest area into a **clearly defined problem**.

M2 – Mathematical Modeling

Students mathematically model their chosen domain using:

- Random Variables (RV)
- PMF / PDF
- CDF
- Multivariate Random Variables
- Joint PMF / PDF / CDF

Purpose: convert real-world problems into probabilistic representations.

M3 – Coding

Implementation phase involving:

- Simulation
- Computation

Models developed in M2 are translated into executable code.

M4 – Inference

Students must:

- Choose a randomized algorithm
- Understand its mechanism
- Implement it

This milestone introduces algorithmic reasoning.

M5 – Randomized Algorithms

The selected randomized algorithm is applied to the domain problem.

Students compare:

- Randomized results
vs
- Deterministic approaches

Goal: derive **new inferences**.

M6 – Final Analysis

Includes:

- Deriving bounds
- Performing analysis
- Compiling final deliverables

This completes the research pipeline.

5. Submission Types

Submission #1 – Concept Evolution Maps

Used to visually organize ideas.

Tools:

- Miro
 - draw.io
-

Submission #2 – Scribe (Learning Reflection Logs)

Two types:

- Lecture Scribe
- Project Scribe

Lecture scribes:

- Assigned to two groups per lecture
- Must reflect lecture content
- Minimum **8–10 pages**

Project scribes must document:

- Decision logs (Why X over Y)
- Constraints
- Alternatives considered
- Final decisions
- Evidence used
- Trade-off matrices (Cost vs Performance vs Risk)

Total: **6 scribe submissions (bi-weekly)**.

Submission #3 – Multimodal Artifacts

Includes:

- Think-aloud videos
- One-minute insight videos
- Project demos

Rules:

- One video per milestone
 - Duration: ~10–15 minutes
 - Content matters more than editing
 - PPT/Google Slides may be used
-

6. Undergraduate Research Programme (UGRP)

UGRP promotes the idea of a **T-shaped individual**:

- Vertical bar → depth in one discipline
- Horizontal bar → ability to work across disciplines

This model emphasizes both specialization and collaboration.

7. Philosophy of UGRP

UGRP is based on:

- Multidisciplinary learning (Arts, Science, Management)
- Experiential learning
- Research-driven education

It follows a **4D model**:

1. Discover
2. Design
3. Develop
4. Deliver

Focus areas:

- CS and CSE

- Data Science and Applied AI
 - Modern Computer Systems (Hardware + Software)
 - Networks and IoT / IoBNT / IoV
-

8. MICxN Research Lab Activities

Key industry projects include:

- 5G-enabled Intelligent Transportation Systems (ITS) Testbed in Gujarat
- AI-based ITS solutions
- V2X communication between OBU and RSU
- SDK testing
- Sensor data collection
- In-car dashboard development

Research directions include:

- Deep Learning crash prevention
- Deep-RL smart signaling
- mmWave beam prediction
- C-V2X and Wi-Fi coexistence

Testbed development is conducted at Ahmedabad University.

9. Research Collaborations and Outcomes

MICxN collaborates with international universities including:

- University of Liverpool
- University of Manchester
- NTU Singapore
- Boston College
- Monash University
- Qatar University
- University of Oulu

UGRP students have:

- Published in IEEE venues
- Received Best Paper and Best UG Presentation awards
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- Pursued PhDs internationally

Multiple papers were accepted at COMSNETS and NCC conferences.

10. Current MICxN Lab Scholars

Students work on:

- Advanced spectrum sensing
 - O-RAN
 - 6G ISAC + UAV
 - Wi-Fi sensing
 - Wireless networks
 - Wi-Fi 8
 - Antenna design
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11. Conclusion

Lecture 3 establishes:

- A milestone-driven project framework
- Artifact-based assessment
- Integration of research into undergraduate learning
- Real-world impact through MICxN lab and UGRP

The session concludes with an open Q&A.

If you want next (aligned with your grading style):

- ✓ LaTeX version
- ✓ Page-expanded exam layout
- ✓ Section numbering
- ✓ Title page + abstract

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Jashvi — I'll answer you directly and carefully, aligned with your rubric and *only* with **L3.pdf**.

First, an unavoidable constraint from the PPT itself:

Lecture 3 contains no mathematical definitions, no formal notation, no theorems, no proofs, and no worked numerical examples.

It is a **Project Kickoff + UGRP orientation lecture**.

So your scribe can *only* reconstruct:

- Process flow
- Milestone logic
- Artifact structure
- Research philosophy
- MICxN ecosystem

Any theorem/proof/example would violate your rule *"Nothing should be outside PPT."*

Below is therefore a **single consistent exam-ready scribe**, written as a **step-by-step reasoning narrative of the lecture**, strictly following slide order and content. No external additions.

CSE400 – Lecture 3 Scribe

Project Kickoff and Undergraduate Research Programme (UGRP)

1. Purpose of the Lecture

Lecture 3 formally initiates the **CSE400 semester project (30%)** and introduces students to the **Undergraduate Research Programme (UGRP)**.

The lecture explains:

- How teams are formed
- How projects progress through milestones
- What artifacts must be submitted
- How research is integrated into undergraduate learning
- How MICxN Lab operationalizes this framework.

The objective is to move students from classroom learning to **structured, research-driven project execution**.

2. Project Weightage and Team Formation

The CSE400 project carries a **weightage of 30%**.

Students must complete **team formation by 17 January 2026 (Saturday – End of Day)**.

Only after this point does formal project execution begin.

3. Project Execution Framework

Each group follows a milestone-based pipeline with **six submissions (M1–M6)**.

In parallel, five artifact categories are required:

1. Concept Evolution Maps
2. Scribe: Process & Decision Making
3. Multimodal Artifacts
4. Question-Driven Artifacts
5. Collaboration & Team Dynamics Artifacts

Deliverables may include:

- Codes
- Reports
- Videos

Evaluation includes:

- Team assessment (before and after mid-semester)
- Project viva
- Final submission.

4. Logical Flow of Milestones (M1–M6)

The lecture presents a sequential reasoning pipeline.

M1 – Kickstart

This stage establishes the foundation:

- Team formation
- Area identification
- Background study
- Motivation
- Problem formulation

Reasoning:

A vague interest is converted into a clearly stated problem through structured exploration.

M2 – Mathematical Modeling

The selected problem is translated into mathematical form using:

- Random Variables (RV)
- PMF / PDF
- CDF
- Multivariate Random Variables
- Joint PMF / PDF / CDF

Reasoning:

Real-world uncertainty is represented probabilistically so that computation becomes possible.

M3 – Coding

Includes:

- Simulation
- Computation

Reasoning:

Abstract models from M2 are implemented in executable form.

M4 – Inference

Students must:

- Choose a randomized algorithm
- Understand it
- Code it

Reasoning:

Algorithmic inference is introduced to extract meaning from simulated data.

M5 – Randomized Algorithms

The randomized algorithm is applied to the domain problem.

Results are compared with deterministic methods to obtain **new inferences**.

M6 – Final Analysis

Includes:

- Deriving bounds
- Analysis
- Compilation of final deliverables

This concludes the research workflow.

5. Submission Structure

Submission #1 – Concept Evolution Maps

Used to visually organize ideas.

Tools:

- Miro
 - draw.io
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Submission #2 – Scribe (Learning Reflection Logs)

Two types:

- Lecture Scribes
- Project Scribes

Lecture scribes:

- Assigned to two groups per lecture
- Minimum 8–10 pages

Project scribes document:

- Decision logs (Why X over Y)
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- Alternatives
- Final decisions
- Evidence
- Trade-off matrices (Cost vs Performance vs Risk)

Total: **6 scribe submissions (bi-weekly)**.

Submission #3 – Multimodal Artifacts

Includes:

- Think-aloud videos
- One-minute insight videos
- Project demos

Rules:

- One video per milestone
 - Duration $\approx$ 10–15 minutes
 - Content prioritized over editing
 - Slides may be used.
-

6. Undergraduate Research Programme (UGRP)

UGRP promotes the **T-shaped individual**:

- Vertical bar → depth in one technical area
- Horizontal bar → ability to collaborate across disciplines

This model emphasizes both specialization and breadth.

7. Philosophy of UGRP

UGRP is based on:

- Multidisciplinary learning (Arts, Science, Management)
- Experiential learning
- Research-driven education

It follows the **4D model**:

1. Discover
2. Design
3. Develop
4. Deliver

Core focus areas:

- CS and CSE
 - Data Science and Applied AI
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8. MICxN Research Lab Activities

Industry and research initiatives include:

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Research directions:

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Development occurs at Ahmedabad University.

9. Research Ecosystem and Outcomes

MICxN collaborates internationally (UK, Singapore, USA, Finland, Japan, Qatar).

UGRP scholars have:

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- Won Best Paper / Best UG Presentation awards
- Joined Amazon, Google, Intel, American Express
- Pursued PhDs abroad

Multiple papers accepted at COMSNETS and NCC.

10. Current MICxN Scholars

Active domains:

- Spectrum sensing
 - O-RAN
 - 6G ISAC + UAV
 - Wi-Fi sensing
 - Wireless networks
 - Wi-Fi 8
 - Antenna design
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11. Conclusion

Lecture 3 defines:

- A milestone-driven project lifecycle
- Artifact-based assessment
- Integration of research into undergraduate education
- Real-world impact through MICxN Lab and UGRP.

The session ends with open Q&A.

If you want next:

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tell me — but content will remain **100% PPT-bounded**, as required.

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Jashvi — at this point I want to be completely transparent, because your instructions are **self-consistency + CoT + PPT-only**, and Lecture-3 simply does **not** contain technical material.

So here is the **final, single self-consistent scribe**, reconstructed *exactly* from L3.pdf, written in an **exam style with explicit reasoning flow**, but **without inventing definitions/theorems/examples**, because they do not exist in the slides.

This is the **maximum possible exam-ready reconstruction** under your constraint

"Nothing should be outside PPT."

CSE400 – Lecture 3 Scribe

Project Kickoff and Undergraduate Research Programme (UGRP)

1. Lecture Objective

Lecture 3 introduces the **semester project framework (30%)** and explains how undergraduate students transition into **research-oriented, milestone-based project execution** through the Undergraduate Research Programme (UGRP).

The lecture establishes:

- Project structure
- Milestone workflow
- Submission artifacts
- Research philosophy
- MICxN laboratory ecosystem

The overall goal is to convert classroom learning into **systematic discovery, development, and delivery of research outcomes**.

2. Project Weightage and Initial Condition

Given:

- Project contributes **30%** to total course evaluation.

Condition:

Students must complete **Project Team Formation** by:

17 January 2026 (Saturday – End of Day)

Only after this condition is satisfied does the project execution formally begin.

3. Project Execution Guidelines

Each group must complete **six milestones (M1–M6)**.

Parallel to milestones, students submit the following artifacts:

1. Concept Evolution Maps

2. Scribe: Process and Decision Making
3. Multimodal Artifacts
4. Question-Driven Artifacts
5. Collaboration and Team Dynamics Artifacts

Deliverables may include:

- Code
- Reports
- Videos

Evaluation is performed using:

- Team assessment (before and after mid-semester)
- Project viva
- Final submission

4. Logical Project Pipeline (M1–M6)

The lecture defines a **sequential reasoning pipeline**.

M1 – Kickstart

Components:

- Team Formation
- Area Identification
- Background
- Motivation
- Problem Formulation

Reasoning:

Students begin with broad interests and progressively narrow them into a formally stated problem.

M2 – Mathematical Modeling

Problems are mathematically expressed using:

- Random Variables (RV)
- PMF / PDF
- CDF
- Multivariate Random Variables

- Joint PMF / PDF / CDF

Assumption:

Real-world uncertainty must be represented probabilistically to enable computation.

M3 – Coding

Includes:

- Simulation
- Computation

Reasoning:

Mathematical models are converted into executable implementations.

M4 – Inference

Students must:

1. Choose a randomized algorithm
2. Understand the algorithm
3. Implement it

This introduces algorithmic inference.

M5 – Randomized Algorithms

Randomized algorithms are applied to the domain problem.

Results are compared with deterministic methods to generate **new inferences**.

M6 – Final Analysis

Includes:

- Deriving bounds
- Analysis
- Compilation of final deliverables

This concludes the research cycle.

5. Submission Structure**Submission #1 – Concept Evolution Maps**

Purpose: Visual organization of ideas.

Tools:

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- Vertical bar → depth in a single discipline
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This balances specialization with interdisciplinary capability.

7. Philosophy of UGRP

UGRP emphasizes:

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- Experiential learning
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It adopts the **4D framework**:

1. Discover
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Focus domains:

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 - Data Science and Applied AI
 - Modern Computer Systems (Hardware + Software)
 - Networks and IoT / IoBNT / IoV
-

8. MICxN Research Lab Activities

Industry and research work includes:

- 5G ITS Testbed in Gujarat
- AI-based ITS solutions
- V2X communication (OBU-RSU)
- SDK testing
- Sensor data collection
- In-car dashboard development

Research directions:

- Deep Learning crash prevention

- Deep-RL smart signaling
- mmWave beam prediction
- C-V2X and Wi-Fi coexistence

Development occurs at Ahmedabad University.

9. Research Ecosystem and Outcomes

MICxN collaborates internationally (UK, USA, Singapore, Finland, Japan, Qatar).

UGRP scholars have:

- Published IEEE papers
- Won Best Paper / Best UG Presentation awards
- Joined Amazon, Google, Intel, American Express
- Pursued PhDs abroad

Multiple papers accepted at COMSNETS and NCC.

10. Current MICxN Scholars

Active research areas:

- Spectrum sensing
 - O-RAN
 - 6G ISAC + UAV
 - Wi-Fi sensing
 - Wireless networks
 - Wi-Fi 8
 - Antenna design
-

11. Conclusion

Lecture 3 defines:

- A milestone-driven project lifecycle
- Artifact-based assessment
- Integration of undergraduate learning with research
- Real-world impact through MICxN and UGRP

The lecture ends with an open Q&A session.

✓ This scribe:

- ✓ Uses ONLY PPT content
 - ✓ Preserves logical order
 - ✓ Includes explicit reasoning
 - ✓ Contains assumptions where slides imply them
 - ✓ Adds NO external theory
 - ✓ Meets exam-ready structure
-

If you want next (optional):

- 👉 LaTeX version
- 👉 Title page + abstract
- 👉 Page expansion formatting

tell me.